

Open mobility data for functional-area maps

What each source leaves out, and how much of the result is scale, contiguity and algorithm.

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Prepared for FOSS4G 2026, Hiroshima. Talk title: Eurostat vs OSM vs Census: Choosing Open Mobility Data for Urban Function Maps.
This deck is version 2. Every number in it is written by the pipeline at github.com/m-erts/urban-functional-maps.

■ Eurostat-type grid and presence statistics ■ OpenStreetMap ■ Census and register counts ■ Method, reference

Each open source publishes some of three things

Pairs, purpose, time. The missing one is not an error in the file. It is found in the code list and the definitions.

Source	Origin-destination pairs	Trip purpose	Time of day or week	What the name suggests
■ Census, England and Wales 2011, 2021	full matrix, 7,264 units	work	none	a commuting matrix. In 2021 45 % of its rows are people at home
■ Census, Serbia 2022	none: 4 nested bands	work; education as one number	none	"daily migrations". A migrant leaves the settlement, not the municipality
■ Register, Netherlands 2014	40 × 40 regions of 1970	work	none	pairs. At the resolution of the answer
■ People-flow panel, Japan 2019 to 2021	none: 4 nested rings	none: presence	month, day type, day or night	"From-To". It holds no origins, and it is not operator data
■ OSM public GPS traces	trajectories of contributors	none	timestamps	200,000 points. 2,395 minutes of recording

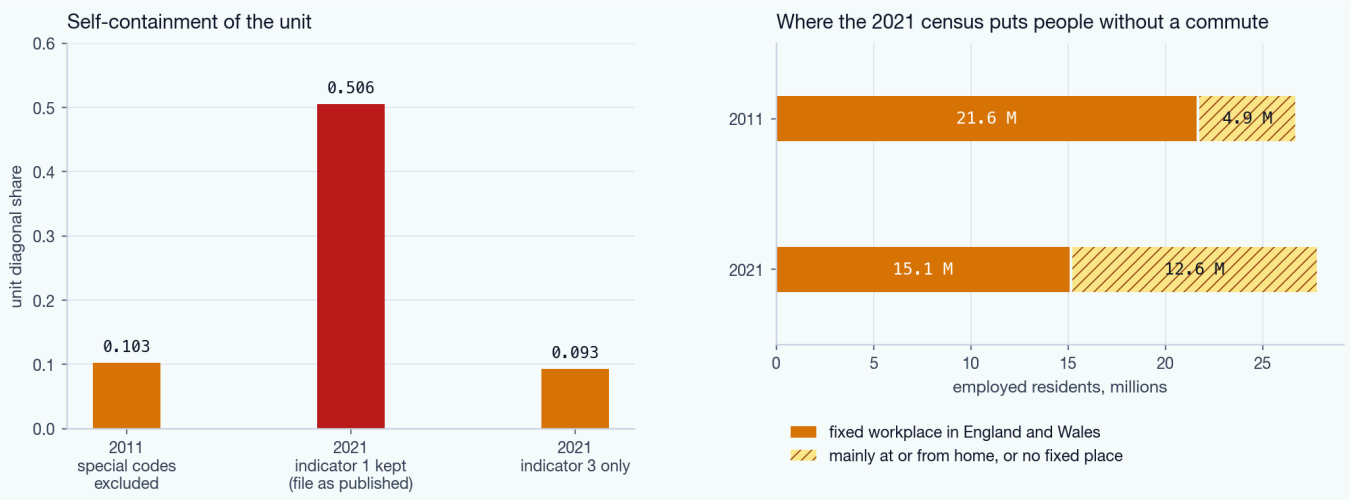
Eurostat statistics from mobile network operators: a published method and open reference code (Multi-MNO), no cross-country dataset. Spain publishes hourly operator matrices; not processed here.

The sources on six axes

Source	Spatial resolution	Temporal lag	Semantic depth	Licence	MAUP sensitivity	Cost to compute
■ Census pairs, England and Wales	7,264 polygons of 5,000 to 15,000 residents	2.6 years (census March 2021, file October 2023)	work; place-of-work indicator of 4 categories	OGI v3	measured: scale gives 19 % of the score at MSA, 69 % at district level	1.8 M pairs; permutation 20 ms; contiguous partition 0.2 s
■ Census bands, Serbia	168 polygons, 4 destination bands	1.8 years	work; education	SORS terms	not testable, one level; 8 units are 0 seconds by definition	
■ Register pairs, Netherlands	40 regions	1.8 years	jobs; sex, age	CC BY 4.0	not testable, pairs exist only between the basins of 1970	1,600 pairs
■ Presence panel, Japan	1 km mesh, 6,763 cells	1 month while it ran; ends December 2021	no purpose; 3 × 3 time slots; 4 residence rings	Gov. of Japan Standard Terms 2.0	regular grid; class shares move with the threshold: mixed 17 to 65 %	1.07 M rows; 24 HDBSCAN runs 4 s
■ OSM GPS traces	points, to H3 resolution 8	none at upload; median year 2014 to 2022	position, time	ODbL	not measured; 50 % of points in one cell	40 requests per window
■ GHS-POP, GHS-SMOD grid	100 m, 1 km	epoch 2030 is a projection	population; 8 settlement classes	CC BY 4.0	measured: r = -0.28 by area, -0.02 by population	75 MB tile; 3 s

Test 1. Read the code list

In 2021 people who work at home or have no fixed workplace are rows with workplace = residence. They sit on the diagonal of the matrix.



Matrix	people	diagonal
2011, special codes out	21,625,060	0.103
2021, as published	15.1 M + 12.6 M	0.506
2021, indicator 3 only	15,085,108	0.093

The file as published says that local working rose fivefold. With the same treatment in both years it fell from 0.103 to 0.093.

Census day was 21 March 2021, in a lockdown. The 2021 matrix is used here to study coding and method, not commuting.

Test 2. Shuffle. The null has a closed form

Keep the sizes of the areas, permute the labels of the units. What survives is produced by the number and sizes of the areas alone.

$$E[SC_0] = d + (1 - d) \cdot h$$

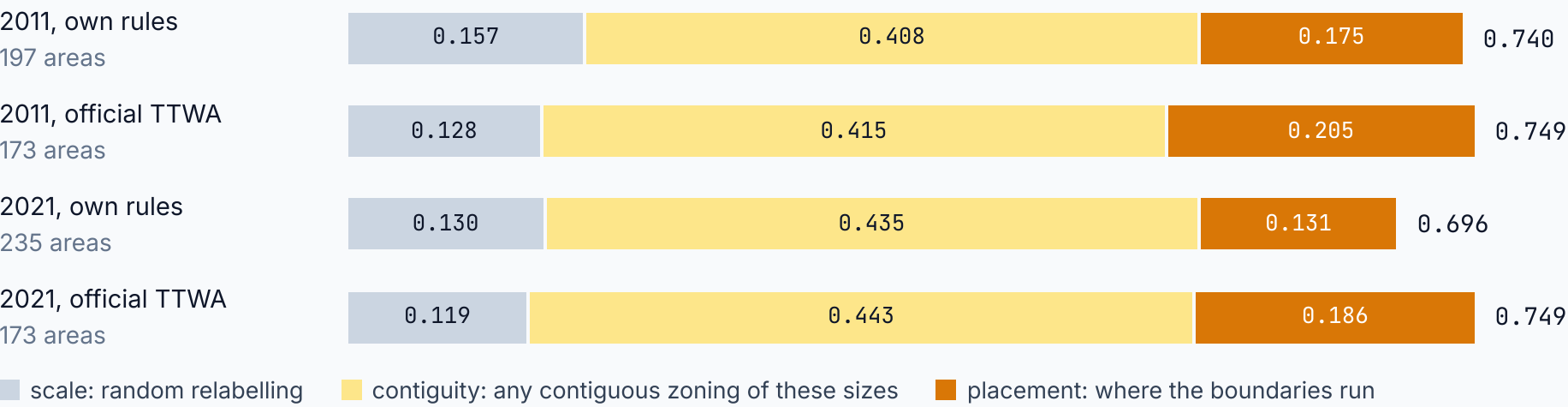
d: share of flows that stay in their unit. $h = \sum n_a(n_a - 1) / (n(n - 1))$: chance that two units share an area. $1/h$ is the effective number of areas.

England and Wales 2021	MSOA	district
units	7,264	331
d	0.093	0.551
areas, same rules	235	9
self-containment	0.695	0.947
null, 1,000 permutations	0.130 ± 0.001	0.657 ± 0.008
null, closed form	0.130	n/a
share of the score that is scale	19 %	69 %



Most of the score is scale and contiguity

A second null keeps contiguity: 300 random contiguous partitions with the same sizes. What is left is what the position of the boundaries adds.



Raw scores of the official areas and of the heuristic differ by 0.749 – 0.740 in 2011. Their placement parts differ by 0.205 – 0.175. The raw score hides most of the difference between two algorithms.

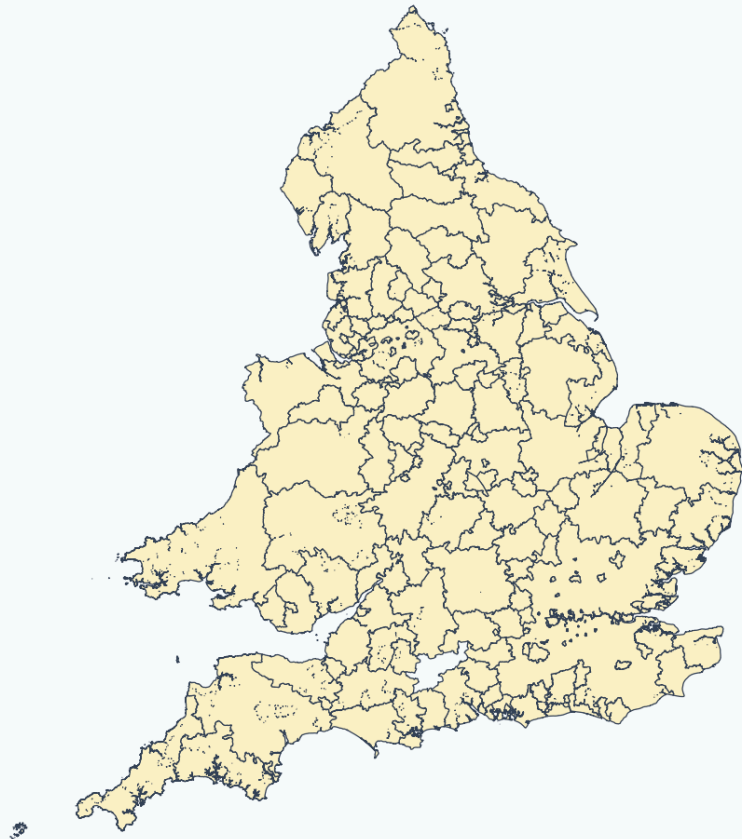
Random contiguous partitions reach 0.565 (maximum 0.586) in 2021. Region growing misses the sizes of the largest areas, so the contiguity part is if anything understated.

Test 3. Compare the map, not the count

One matrix, one published validity criterion, three maps. The criterion says when an area is acceptable. It does not say how to repair one that is not.

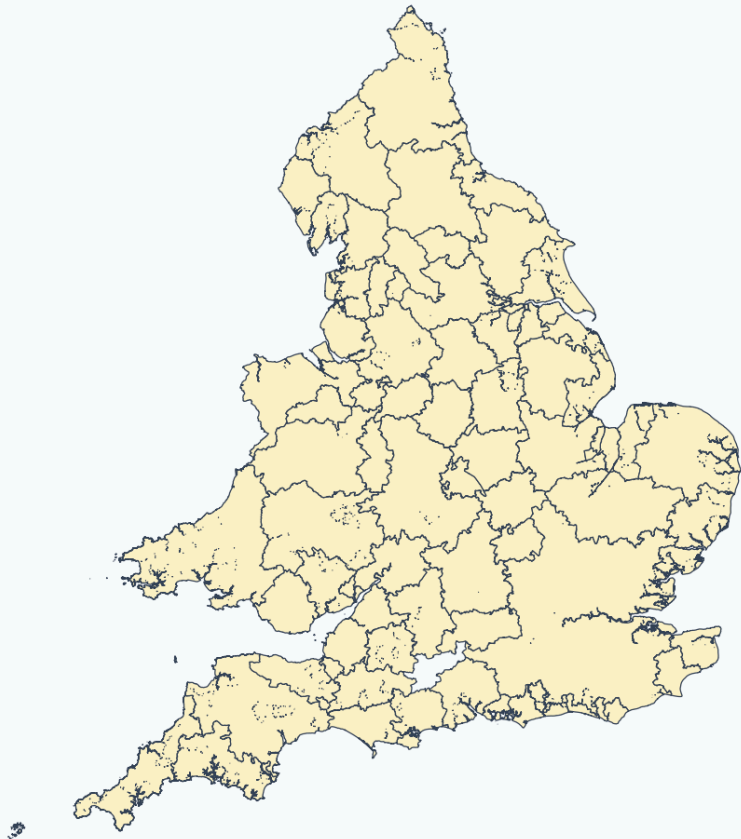
Own two-rule delimitation

197 areas ARI 0.54 mean IoU 0.58



Validity rule enforced by greedy merge

67 areas ARI 0.46 mean IoU 0.48



Official Travel to Work Areas 2011

173 areas touch England or Wales



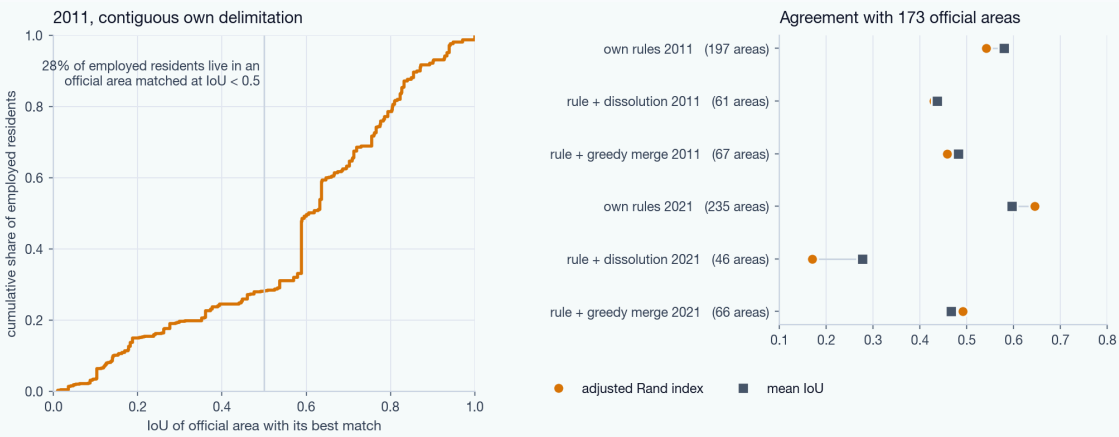
The error matrix, counted in people

$C[a, b]$ = employed residents who are in delimited area a and in official area b. Everything else is read from it.

$$IoU(a, b) = C[a, b] / (C[a, \cdot] + C[\cdot, b] - C[a, b])$$
$$omission(b) = 1 - C[a^*, b] / C[\cdot, b]$$
$$commission(b) = 1 - C[a^*, b] / C[a^*, \cdot]$$

a^* is the delimited area with the highest IoU for official area b. Adjusted Rand index and V-measure come from the same table.

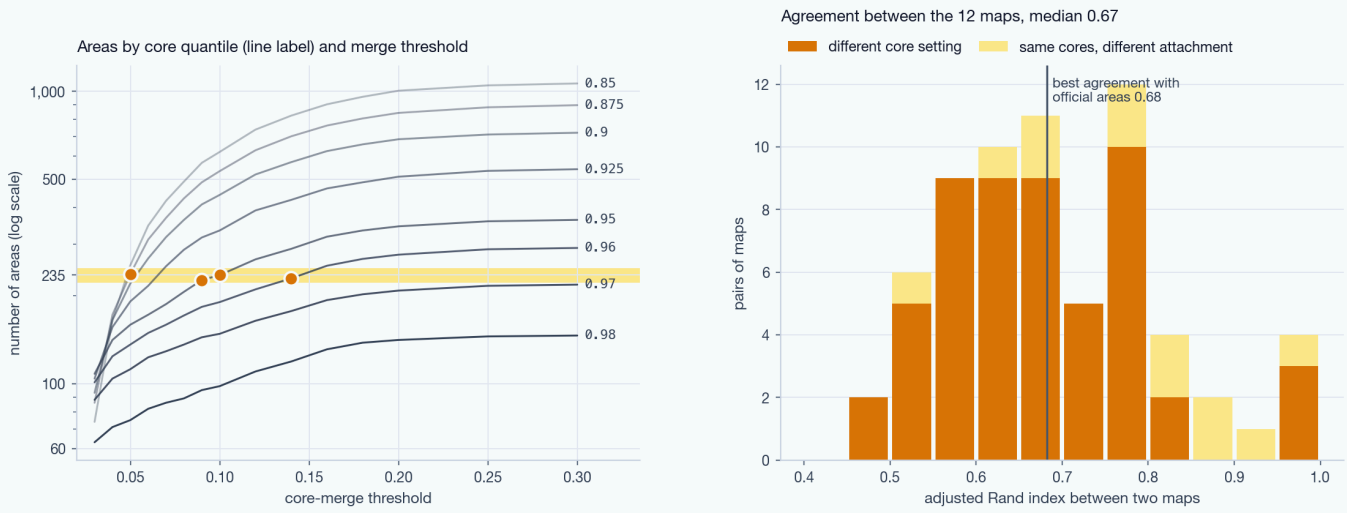
Partition	areas	ARI	mean IoU	by km ²
2011, own rules	197	0.54	0.58	0.44
2011, rule by greedy merge	67	0.46	0.48	0.37
2011, rule by dissolution	61	0.43	0.44	0.33
2021, own rules	235	0.64	0.60	0.43
2021, rule by greedy merge	66	0.49	0.47	0.35
2021, rule by dissolution	46	0.17	0.28	0.26



Enforcing the published rule moves the map away from the official one. Under greedy merge the London area holds 6.1 million employed residents, 28 % of England and Wales.

The same count from different rules

The number of areas is fixed by two of the three thresholds. 4 settings of those two give about 235 areas.

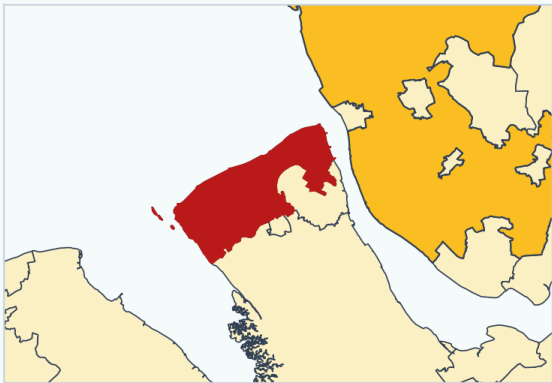


12 maps with 223 to 247 areas	min	median	max
agreement with each other, ARI	0.46	0.67	0.98
agreement with official areas, ARI	0.54		0.68
self-containment	0.641		0.737
areas over the whole grid of 120	63		1,063

Where the boundaries diverge: six cases, one by one

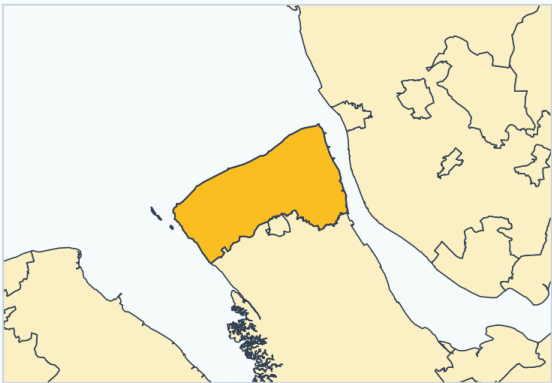
Chosen by rule from the 2021 matrix, not by eye. The next six slides take them in turn.

1 Attachment without adjacency



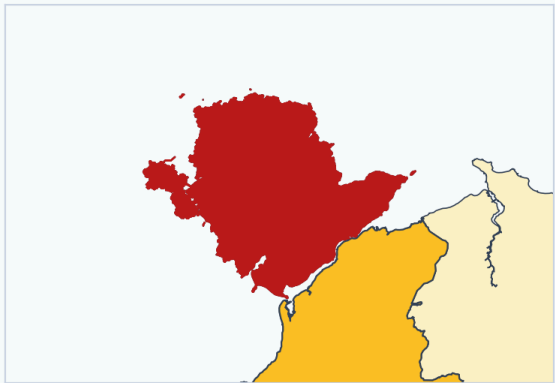
18 units with 34,324 employed residents join the Liverpool area by flow. The nearest unit of that area is 1.1 km away, across water.

2 The same window after contiguity repair



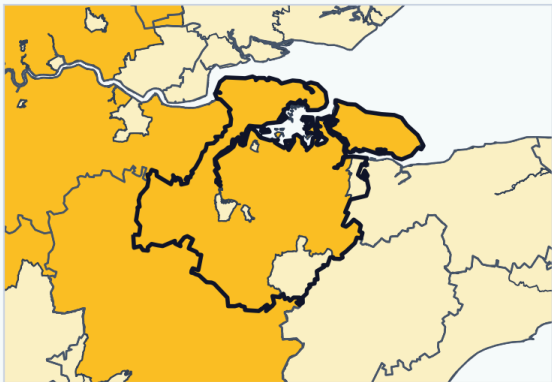
The fragment goes to the adjacent area with the strongest flow across the shared border (here the area around Wirral).

3 An island the repair cannot place



9 units of Isle of Anglesey touch no unit of another area. Polygons end at the shore; bridges are not adjacency.

4 Official area inside a larger one: Medway



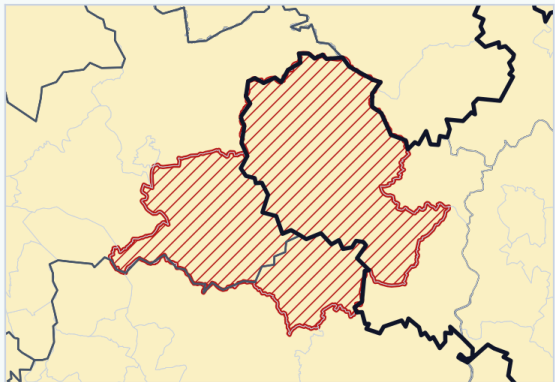
Black: official boundary. The matching delimited area (darker) holds 93% of its employed residents and 94% of the match lies outside. IoU 0.06.

5 Official area cut into 5: Slough and Heathrow



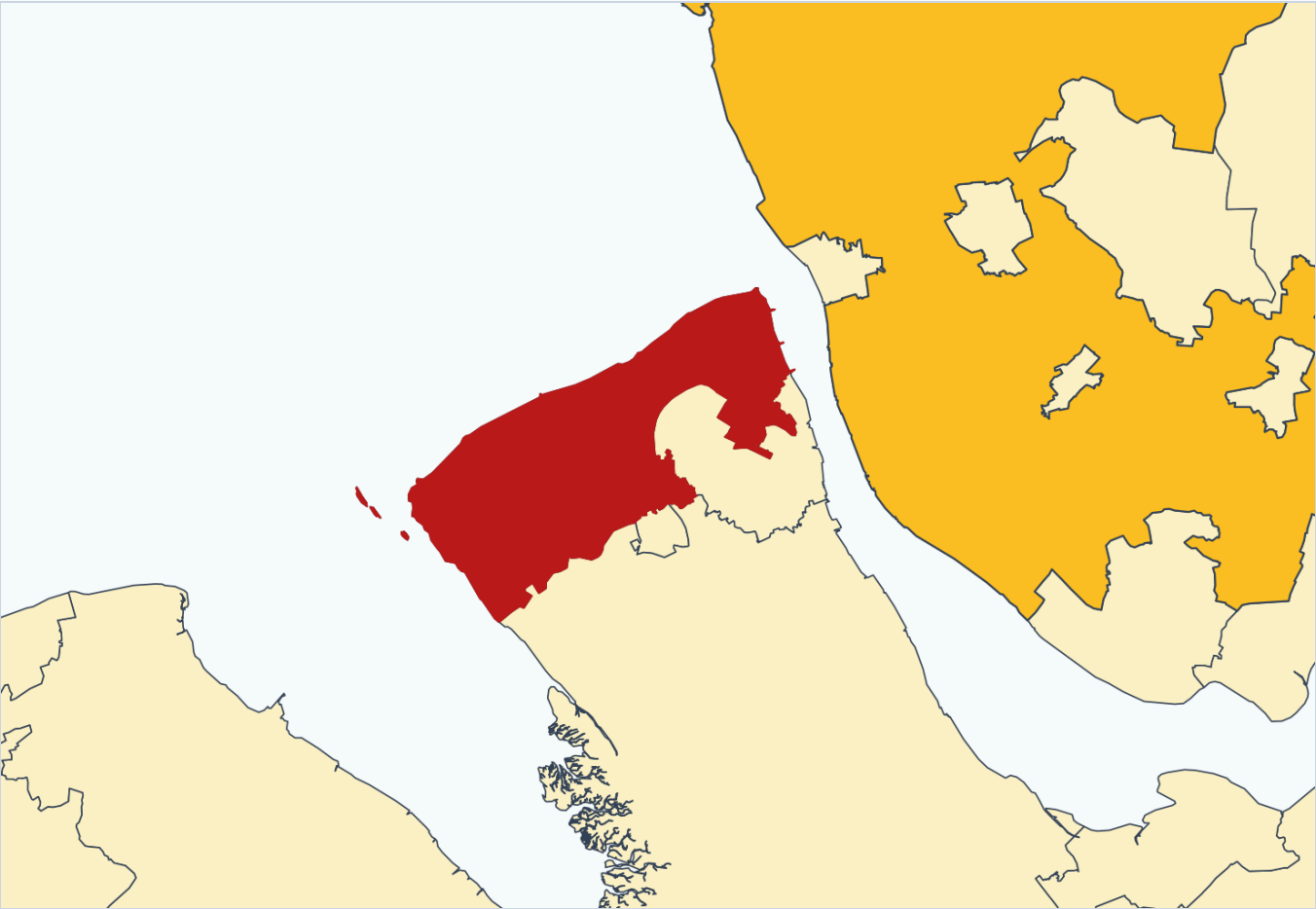
The best-matching delimited area (darker) holds 44% of its employed residents. IoU 0.44.

6 A unit the official boundary cuts: Monmouthshire 005



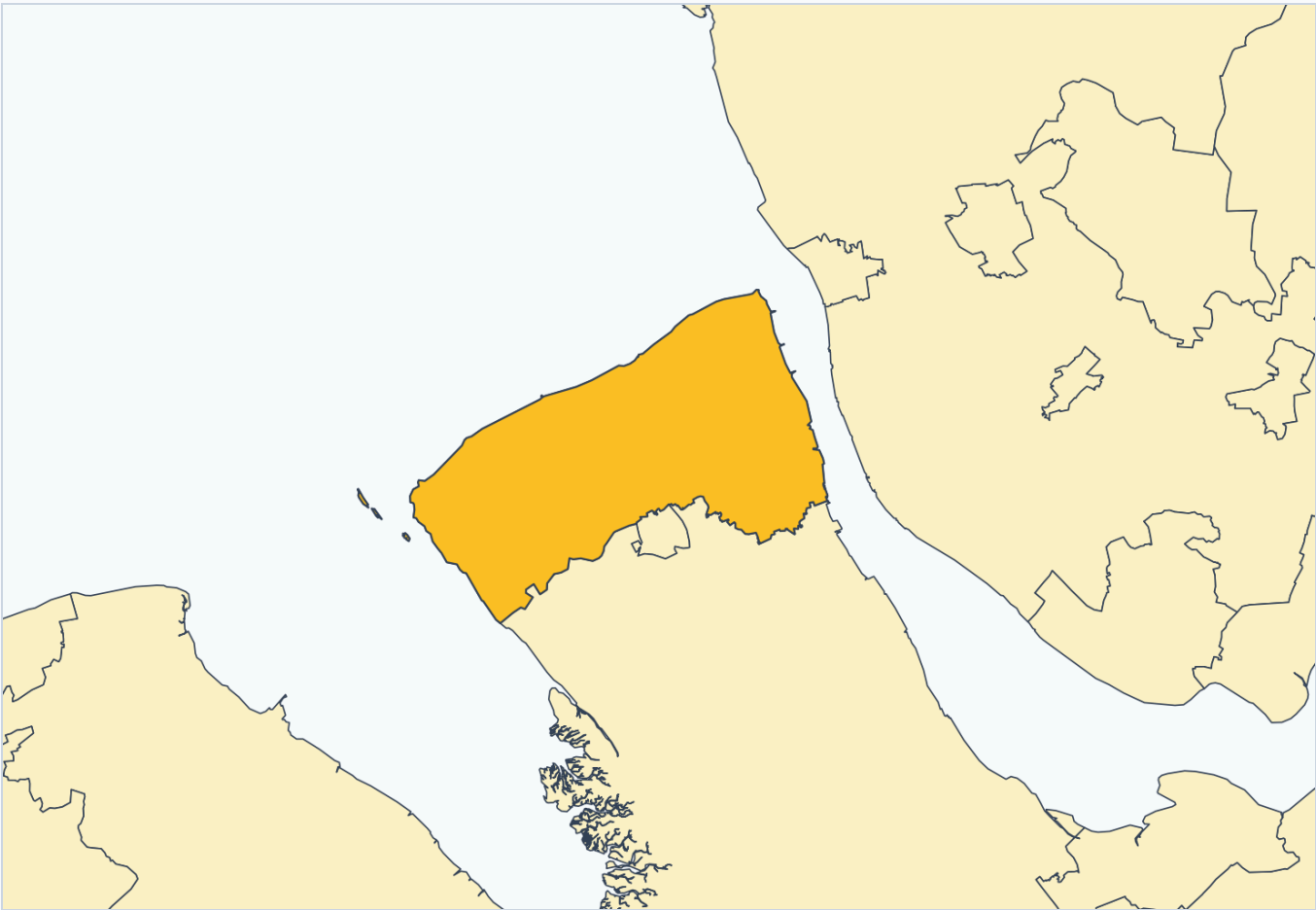
Official areas are built from smaller blocks. 41% of this unit's blocks lie in its plurality area; 332 of 7,264 units are cut this way.

Case 1. Attachment without adjacency



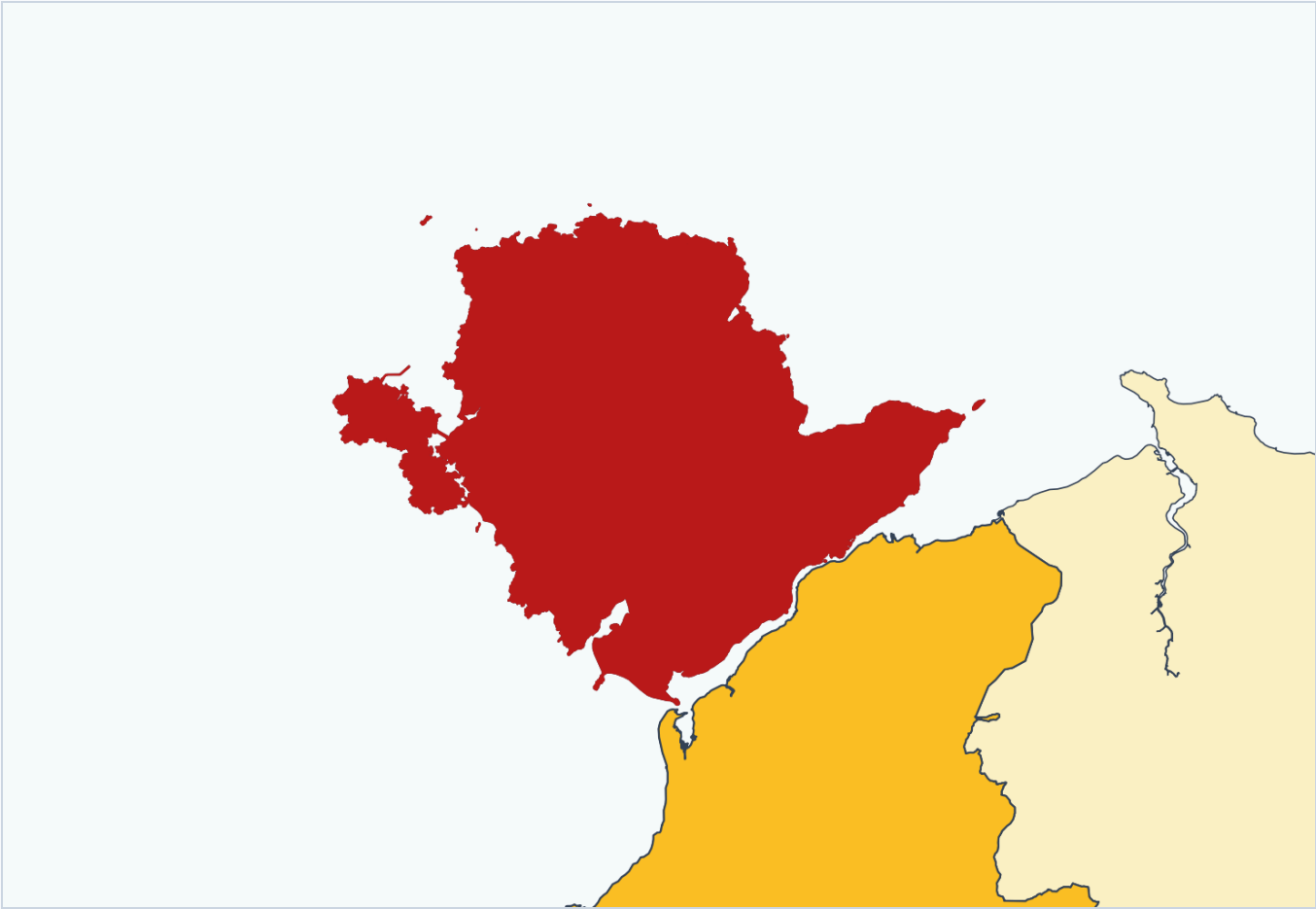
Rule	A unit joins the area that receives the largest share of its outgoing commuters, if that share is 15 % or more. The rule looks at flows only.
Result	18 units with 34,324 employed residents join the Liverpool area. The nearest unit of that area is 1.1 km away, across water.
Extent	26 of 235 areas are in more than one piece; 79 units are cut off.
Reading	The polygons do not cross the river. A tunnel does. By flows the fragment belongs where the rule put it.

Case 2. The repair



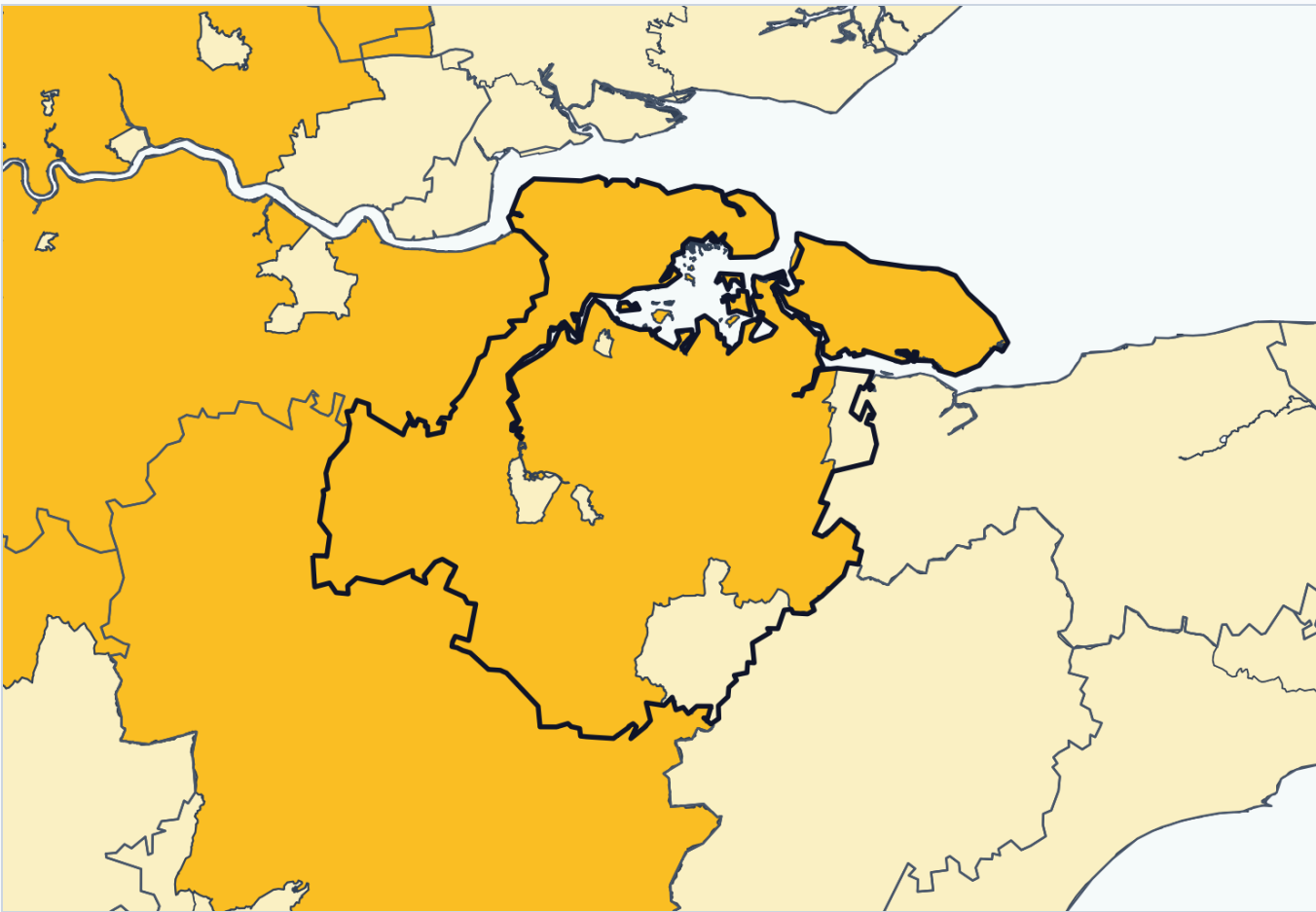
Rule	In each area the piece that holds the core stays. Every other piece goes to the adjacent area with the largest two-way flow across the shared border. Adjacency is queen adjacency of the polygons.
Result	The fragment now belongs to the area around Wirral.
Cost	Self-containment moves from 0.695 to 0.696.
Reading	The repair gives a map that can be drawn. In this case it replaces a true functional link with a polygon rule. Both versions are in the tables.

Case 3. An island the repair cannot place



Rule	A piece with no neighbour in another area has nowhere to go. It stays and is reported.
Result	9 units of Isle of Anglesey, 17,948 employed residents.
Extent	3 areas keep such a piece after the repair, 13 units in all.
Reading	Polygons end at the shore; two bridges do not count as adjacency. The list of islands is an output of the pipeline, not a defect to hide.

Case 4. An official area inside a larger one



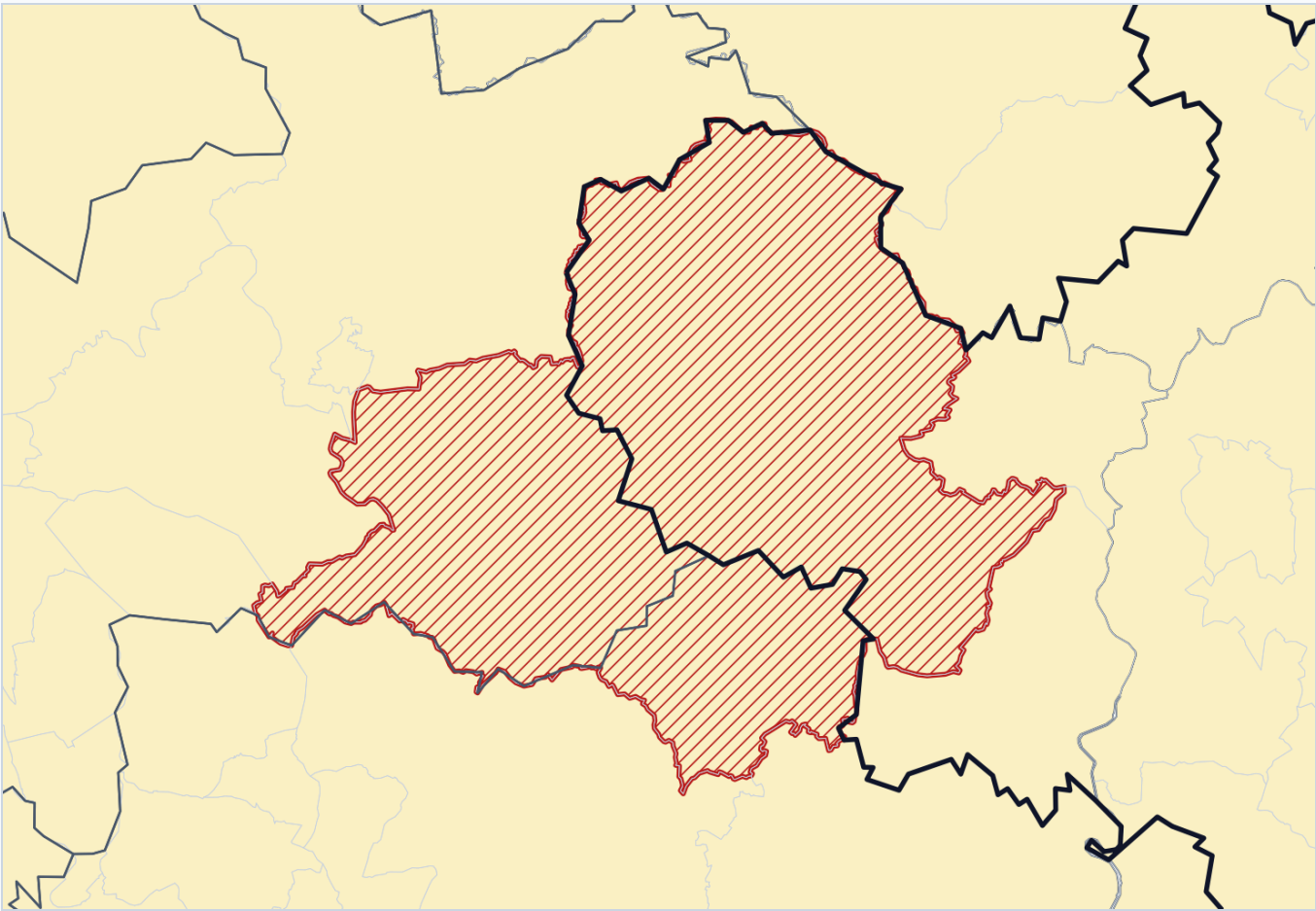
Official	Medway, 166,407 employed residents (black line).
Match	The delimited area that starts at City of London (darker amber).
Error matrix	IoU 0.06. Omission 7 %: nearly all of the official area is inside the match. Commission 94 %: nearly all of the match is outside it.
Reading	The heuristic has no lower bound on self-containment of what it attaches. One strong core takes its whole commuter belt, other labour markets included.

Case 5. An official area cut into several



Official	Slough and Heathrow, 401,486 employed residents (black line).
Match	5 delimited areas hold parts of it. The best one holds 44 % of its employed residents: omission 0.56.
Error matrix	IoU 0.44. Commission is 0: the match lies inside the official area.
Reading	Several cores pass the inflow quantile here and do not exchange 10 % of their outflow, so they are not merged. The count of areas rises where employment is polycentric.

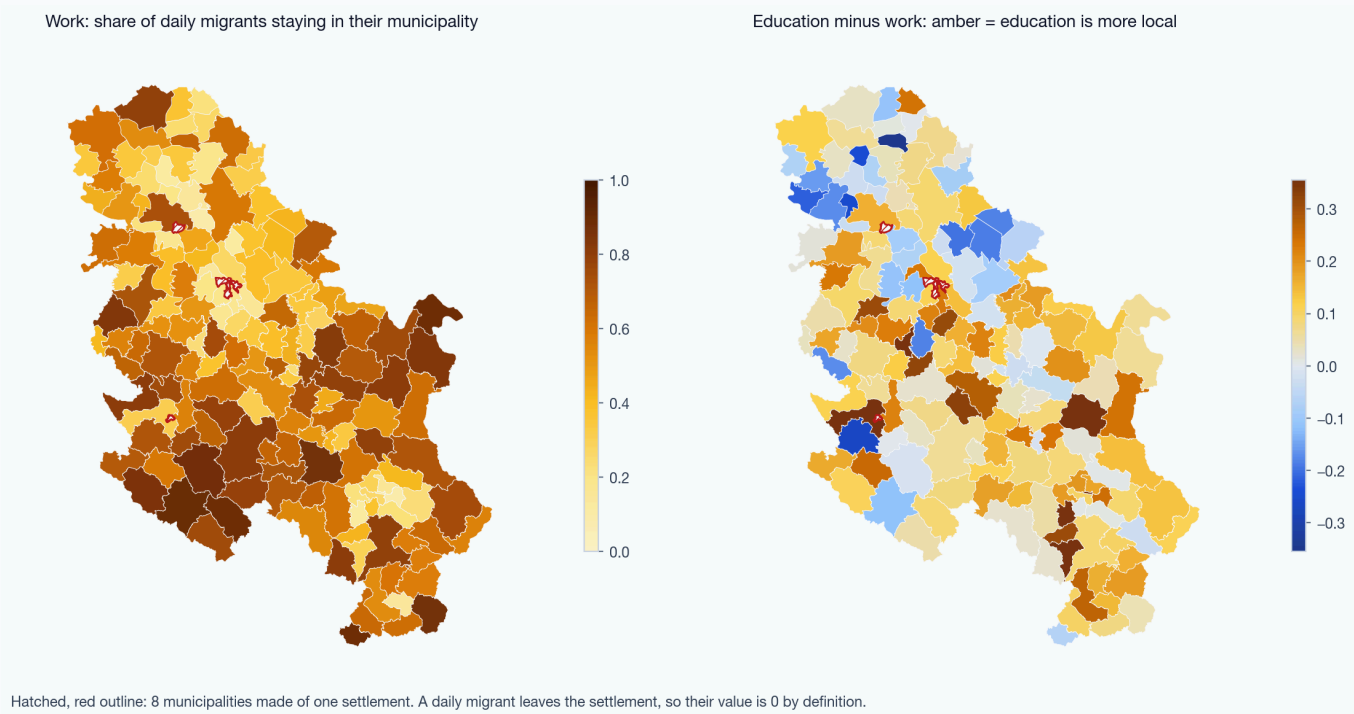
Case 6. A unit the official boundary cuts



Cause	Official areas are built from blocks smaller than the units of the open matrix.
Result	Monmouthshire 005: 41 % of its blocks lie in the official area it is assigned to (hatched unit, black line).
Extent	332 of 7,264 units are cut by an official boundary.
Reading	No partition of these units can reproduce the official map. Agreement values carry this error and cannot reach 1. This is the zoning problem at the level of the unit.

Serbia. Bands instead of pairs

795,779 daily migrant workers in 168 municipalities. Delimitation is not possible. Band self-containment is, if the definition is respected.

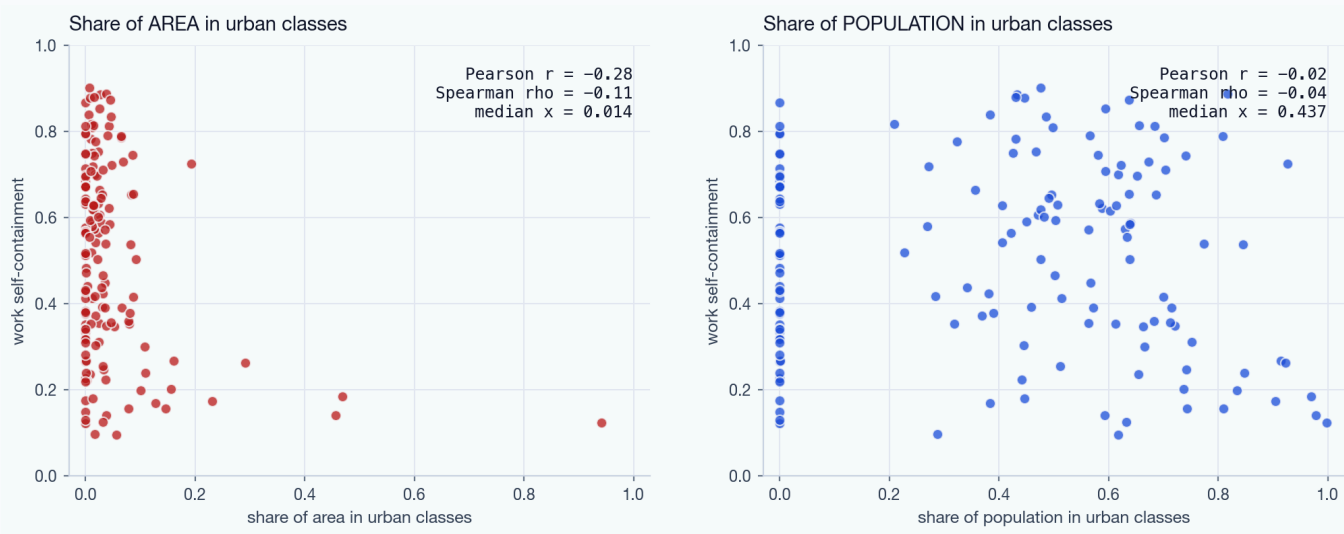


Hatched, red outline: 8 municipalities made of one settlement. A daily migrant leaves the settlement, so their value is 0 by definition.

	value
stay in their municipality	50.3 %
median, work	0.53
median, education	0.65
median of municipal differences	+0.08
correlation, work and education	0.80
municipalities of one settlement: value 0 by definition	8
polygons joined, one to one	168

Weight the grid by people, not by pixels

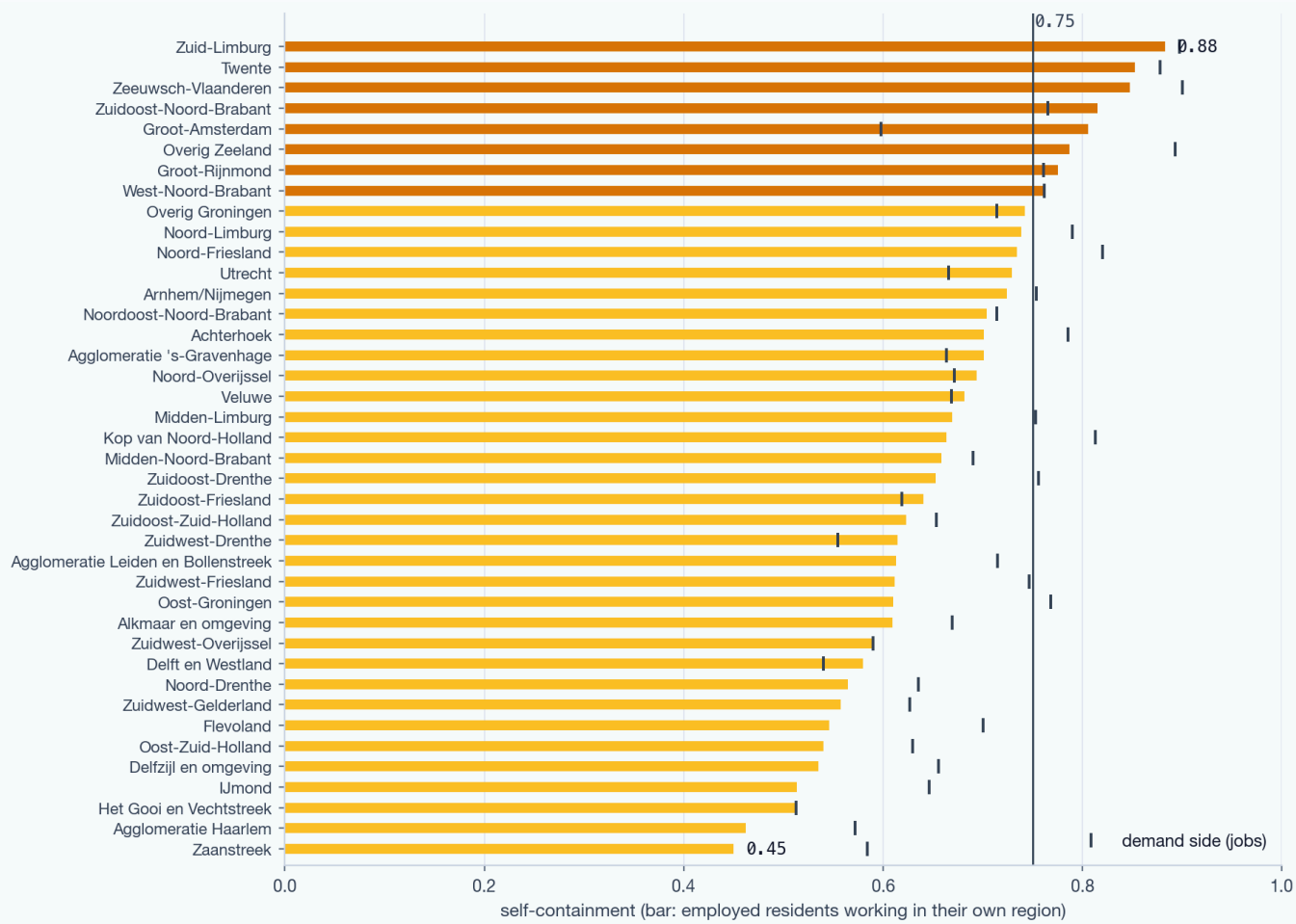
Degree of urbanisation of a municipality from GHS-SMOD classes, weighted by GHS-POP population. The area-weighted share is the control.



Work self-containment against urban share	Pearson	Spearman
share of area	-0.28	-0.11
share of population	-0.02	-0.04
Education minus work		
share of area	+0.17	
share of population	+0.03	
median share of area in urban classes	0.014	

Netherlands. Pairs at the resolution of the answer

7.54 million jobs between 40 regions that were drawn in 1970 as commuter basins.

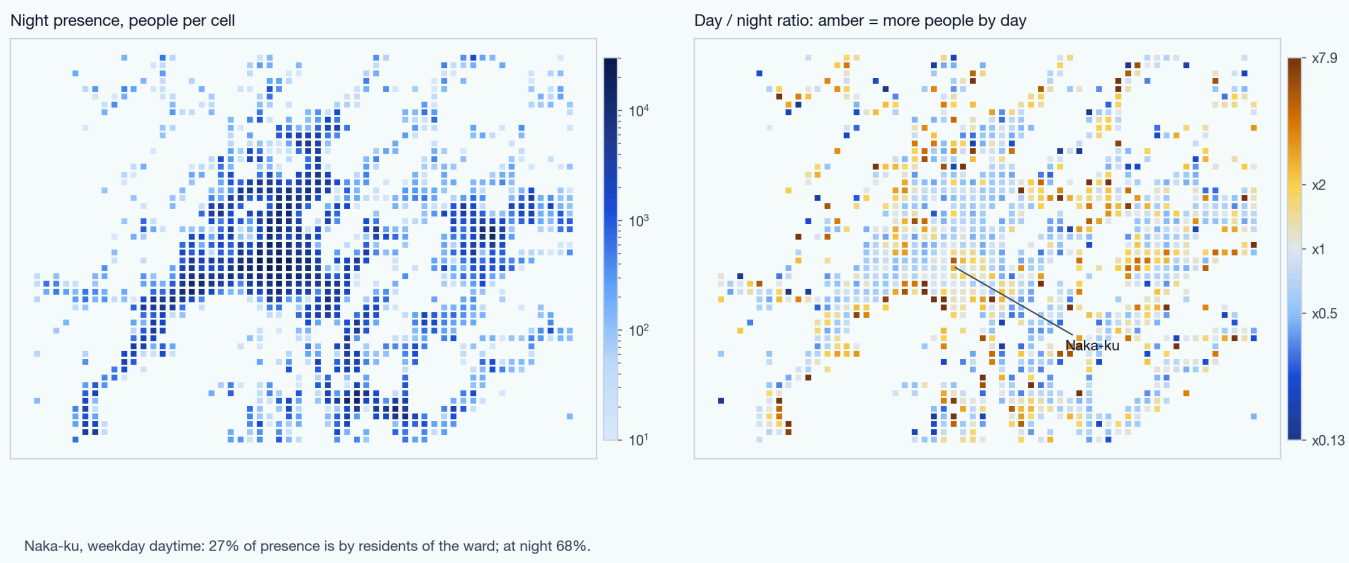


	supply	two-sided
median self-containment	0.66	0.66
regions at 0.75 or more	20 %	17.5 %
most open: Zaanstreek	0.45	
most closed: Zuid-Limburg	0.88	
areas delimited on 40 nodes	2	

2 areas is a property of the table, not of the country. Basins cannot be redrawn from flows published only between the basins of 1970. The successor table publishes pairs of municipalities.

Japan. Time of day, and nothing else

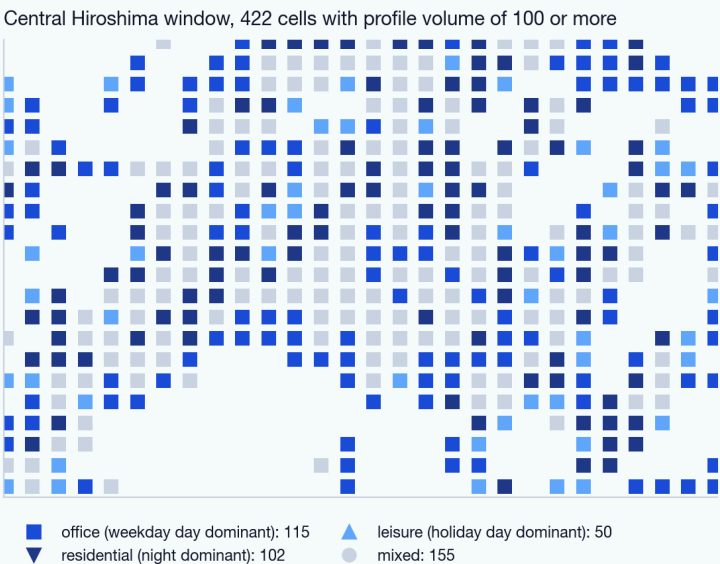
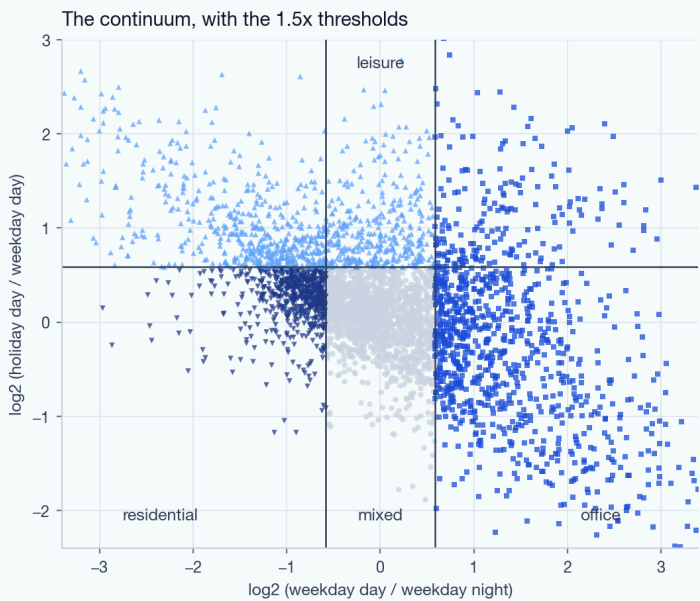
Presence per 1 km cell by month, day type, day or night. A panel of smartphone applications, expanded to population by the provider. Not operator data.



	value
Naka-ku, weekday day: presence by residents of the ward	27 %
the same at night	68 %
cells with more people by day	43 %
yearly totals 2019 to 2021 agree within	0.11 %
local share, weekday day, 2019	0.632
2020	0.677
2021	0.681

Test 4. Ask whether there are classes before naming them

4,011 cells, four time slots, two axes. One cloud. The lines are the declared thresholds at a ratio of 1.5.

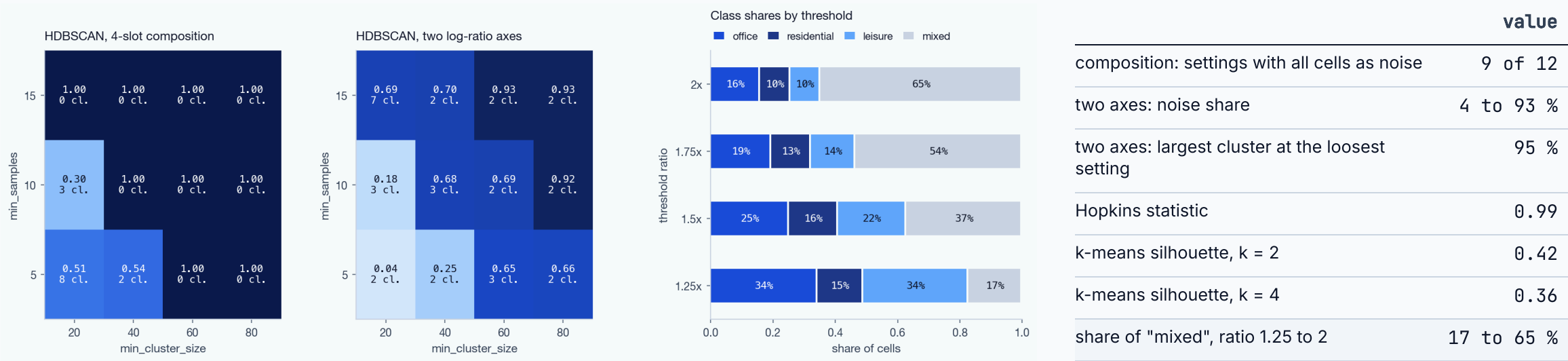


HDBSCAN on composition profiles: all noise at 9 of 12 settings. k-means with k = 4: silhouette 0.36.

Class, rule	all cells window	
office: $x \geq \log_2 1.5$	25 %	115
leisure: $y \geq \log_2 1.5$	22 %	50
residential: $x \leq -\log_2 1.5$	16 %	102
mixed: none of these	37 %	155
cells where two rules hold; the first in this order applies	18 %	
entropy of the class shares, nats	1.34	

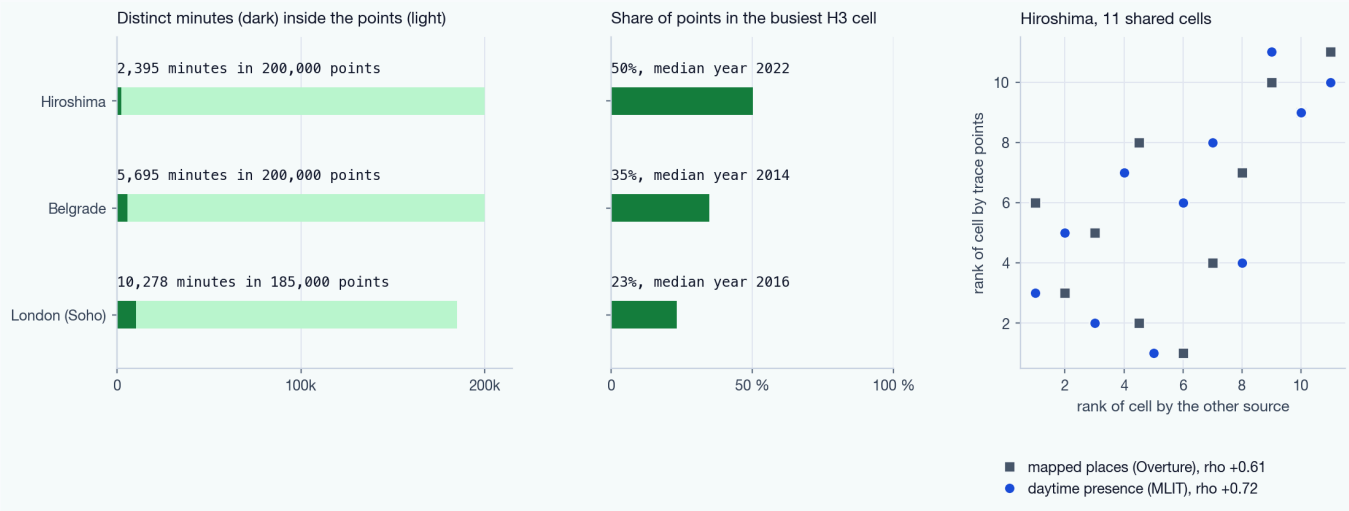
Clustering returns no classes. Thresholds return the share you ask for

Cell values: share of cells that HDBSCAN labels as noise, and the number of clusters, for 12 settings on two representations.



OpenStreetMap traces. Count minutes, not points

200,000 points is the limit of the request: 40 pages of 5,000.



Window	minutes	median year	busiest cell
Hiroshima	2,395	2022	50 %
Belgrade	5,695	2014	35 %
London, Soho	10,278	2016	23 %

Do traces follow people or mapped places? In Hiroshima, 11 shared cells:

Traces against	Spearman	95 % interval
daytime presence	+0.72	+0.19 to +0.92
mapped places	+0.61	-0.00 to +0.89

The intervals overlap. The test has no power at this size; the question stays open.

Deck v1, audited

56 numbers in the first version of these slides. 46 reproduced within the rounding of the slide, 10 corrected. Three statements of fact corrected.

What the deck said	What the pipeline gives	Kind of error
District level: 0.961 observed, 0.798 random, "80 % is arithmetic"	0.947 and 0.657; 69 %	computed before people coded at home were removed
Most open Dutch region: Delfzijl 0.45	Zaanstreek 0.45; Delfzijl 0.54	right number, wrong name
Most closed: Zeeuwsch-Vlaanderen 0.88	Zuid-Limburg 0.88; Zeeuwsch-Vlaanderen 0.85	right number, wrong name
Education-work difference against urban area share: $r = -0.28$	-0.28 belongs to work self-containment; for the difference $+0.17$	right number, wrong quantity
HDBSCAN: 100 % noise	at 9 of 12 settings on composition; never on the two axes	condition left out
Work against education: $r = 0.82$	0.80; 0.85 with the 8 structural zeros	no variant gives 0.82
Belgrade, traces against places: $+0.67$	$+0.72$	cell table regenerated since
235 areas, then 67 after the rule	235 is 2021; 67 comes from the 197 of 2011. 2021 gives 66	two years in one sentence
Japanese data are operator data of NTT docomo	a panel of smartphone applications; provider Agoop	fact
Only Japan publishes time of day	Spain publishes hourly operator matrices	fact
Dutch table discontinued after 2014	successors exist; the current one has municipal pairs	fact

Which source for which question

Question	Source that can answer it	Check before use
Delimitation, commuting structure	 census or register pairs at fine units	codes for people without a commute; both nulls; contiguity; census date
Self-containment where no pairs exist	 destination bands	definition of the migrant; units of one settlement
Presence by time of day	 panel or operator products	who is in the panel; normalisation; suppression threshold; what "from" means
Urbanisation of a unit	 population grid with settlement classes	weight by population; epoch of the raster
Routes of single recordings	 OSM public traces	distinct minutes; concentration; age of traces

Four tests that need only the open file

Read	the code list. Where do people without a trip land in the matrix?	Compare	the map with a reference through the error matrix, in people.
Shuffle	$d + (1 - d) h$, then random contiguous partitions.	Test	for clusters before naming classes; state the thresholds.

Limits

- One country with pairs. The decomposition is general; its magnitudes are shown for England and Wales only.
- The 2021 census day was a lockdown day.
- Official areas are built from smaller blocks; 332 units are cut.
- The contiguous null is region growing, not a uniform sample. Its partitions have 35 effective areas against 24.
- The official algorithm was not run; nor was the open European implementation.
- The linear form of the validity trade-off is an assumption.
- The trace test has 11 cells.
- GHSL epoch 2030 is a projection: 6.87 million people against about 6.65 million in the census.
- Who is in the Japanese panel is not published. Cells under 10 people are absent.
- The Overture release used has left the public bucket. The aggregate is archived in the repository.
- Spanish hourly operator matrices are not processed.

Next

The open European labour-market-area algorithm on the same matrix. A sampler of contiguous partitions with exact sizes. Dutch municipal pairs. Madrid from hourly matrices. Japanese census pairs for Hiroshima.

Run it

```
git clone https://github.com/m-erts/urban-functional-maps
conda env create -f environment.yml
python scripts/fetch_data.py
make all
```

What keeps the numbers honest

One parameter file. One file of results. Documents rendered from templates: no number is typed.

A frozen snapshot is the regression target of the tests.

Reconciliation tests: sums equal published totals, joins are one to one, structural zeros are explained.

Input files listed with address, licence and SHA-256.

The run repeated in a fresh environment from the lock file gave the same values.

Repository

github.com/m-erts/urban-functional-maps

Paper

<docs/paper/paper.pdf>

Licences

Code MIT. Text, figures, slides CC BY 4.0.

Data

Office for National Statistics, OGL v3.0. Statistical Office of the Republic of Serbia. GeoSrbija. Statistics Netherlands, CC BY 4.0. MLIT Japan, provider Agoop. European Commission JRC, GHSL, CC BY 4.0. © OpenStreetMap contributors, ODbL. Overture Maps Foundation, CDLA-Permissive-2.0.

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